Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	2 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Home Page	Display project overview
		Display project features
		Navigate to Prediction page
FR-2	Weather Data Input	Enter weather parameters
		Validate input values
		Submit prediction request
FR-3	Flood Prediction	Preprocess input data
		Load Decision Tree model
		(`decision_tree_flood.pkl`)
FR-4	Prediction Results	Display Flood or No Flood result
		Display safety precautions

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The application should provide a simple, user-friendly, and responsive interface that allows users

		to easily enter weather data and view flood prediction results.
NFR-2	**Security**	The application should validate user inputs, handle errors securely, and protect the machine learning model and application files from unauthorized modification.
NFR-3	**Reliability**	The system should generate consistent and accurate flood predictions using the trained Decision Tree model and operate without failures during normal usage.
NFR-4	**Performance**	The application should process user inputs and display flood prediction results within a few seconds while efficiently utilizing system resources.
NFR-5	**Availability**	The application should be available whenever required and support deployment on a local server, with future cloud deployment support.
NFR-6	**Scalability**	The system should support future enhancements such as real-time weather APIs, cloud deployment, additional machine learning models, and larger datasets without major architectural changes.